



Does Distillation of Large Language Models Preserve Watermarks?

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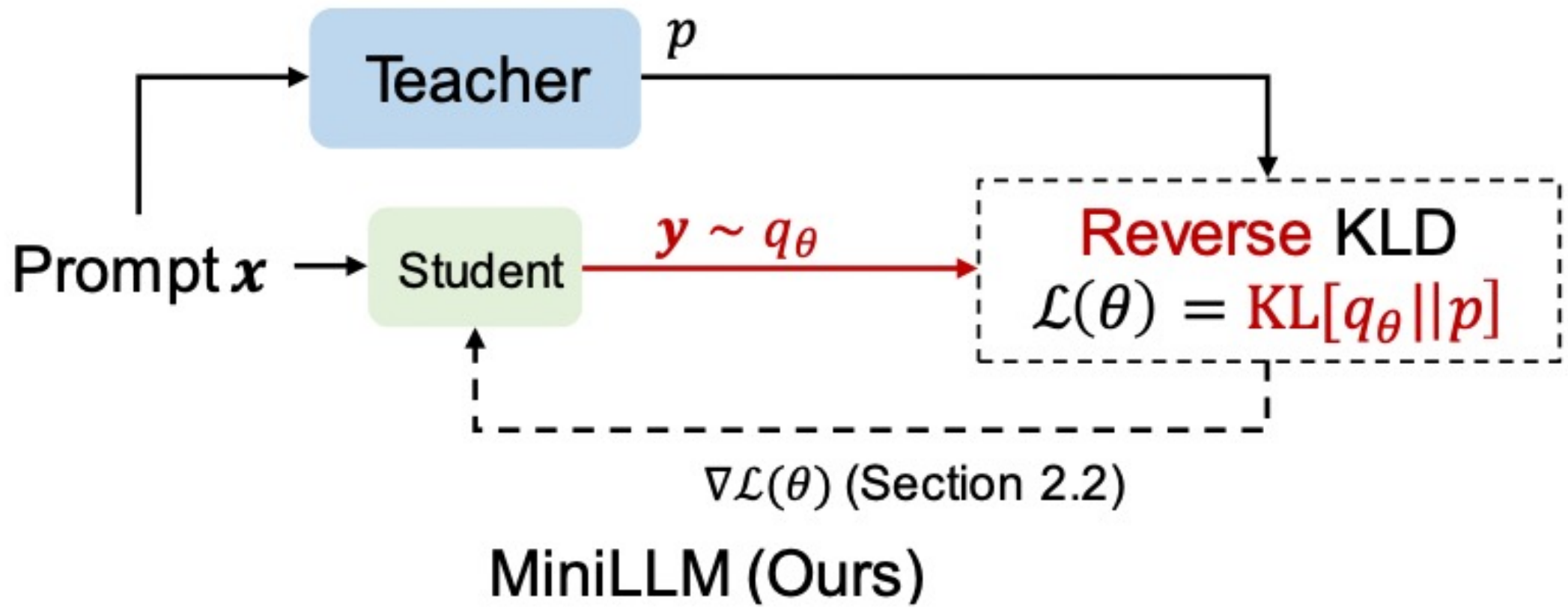


Introduction

- Recent advancements in LLMs have led to active research on watermarking techniques for security and ownership
- These watermarking techniques embed signals that are imperceptible to humans into the model's outputs, allowing for verification if a given output was machine generated
- Simultaneously, to reduce the recourse consumption caused by the increasing size of LLMs, LLM distillation technology has recently gained attention. Distillation involves transferring the knowledge of a large model (teacher model) to a smaller model (student model), effectively reducing the model size and inference time
- The goal of this study is to explore and understand whether watermarked LLM models maintain their original watermarks during the distillation process**
- This study also proposes new algorithms designed to preserve watermarks during the distillation process if not preserved**

Distillation Technique

Figure 1: Our team employs MiniLLM as our knowledge distillation (KD) technique, enabling the student model to effectively retain and apply knowledge from the teacher model through iterative learning



What is KD?: The primary goal is to distill the knowledge from a large teacher model into a smaller student model while maintaining the performance of the larger model.

MiniLLM Optimization: Leverages “**Policy Gradient Theorem**” to compute gradients for updating the student model's parameters, focusing on maximizing rewards that align student outputs with the teacher's distribution. This involves policy optimization techniques that ensure the student model generates high-quality text

MiniLLM Training Algorithm: Involves fine-tuning the student model on a dataset under the supervision of the teacher model

Watermarking Technique

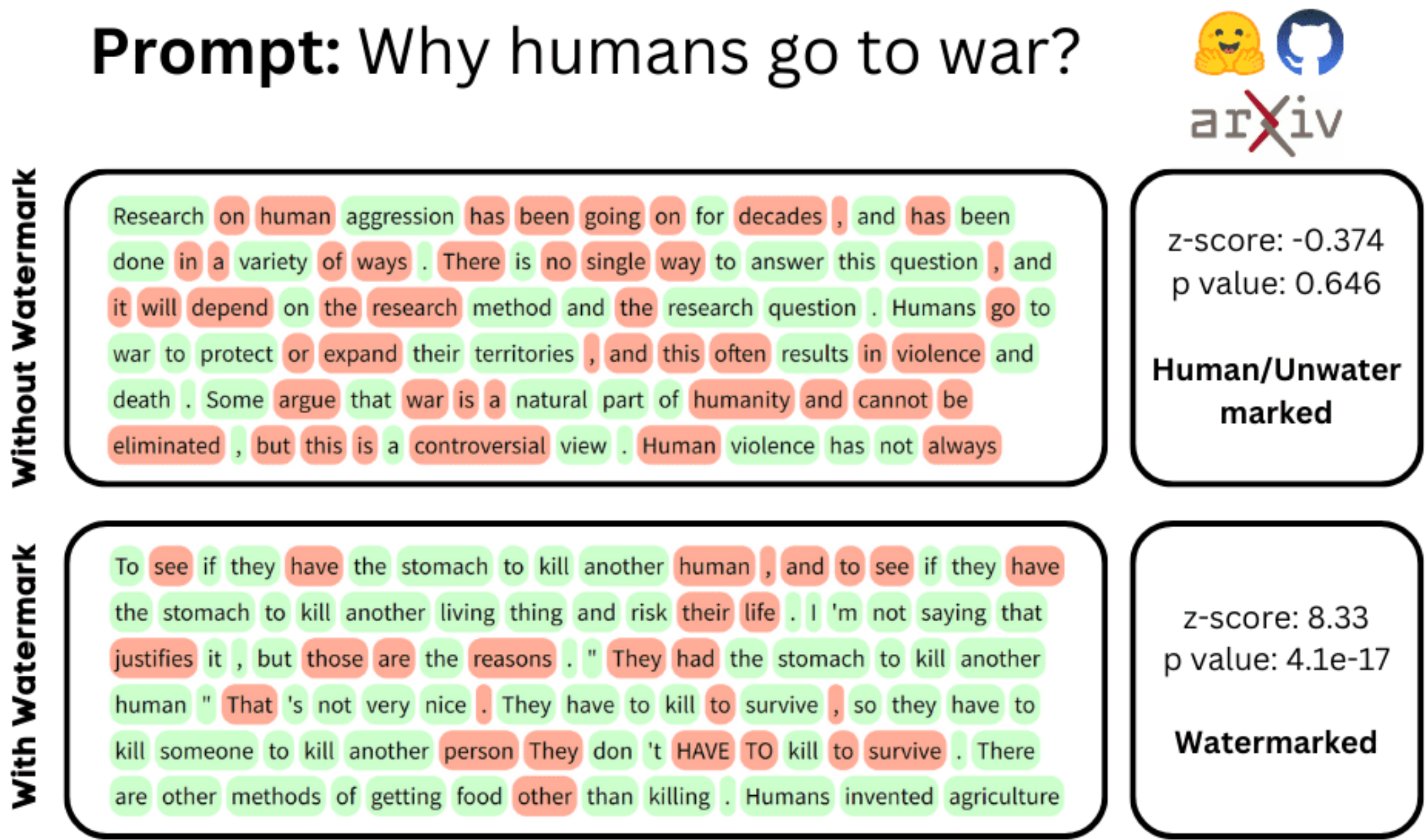


Figure 2: Compares text generated by a large language model (LLM) with and without a watermark. Demonstrates how statistical analysis can distinguish between machine-generated (watermarked) text and unwatermarked text.

What is a Watermark?: A watermark is an invisible pattern embedded within text by a language model. It can be detected algorithmically to identify machine-generated content, by manipulating the distribution of certain words, known as "green tokens" and "red tokens."

Why is the Lower Box Watermarked?: Watermarked due to the increased frequency of "green tokens" and the decreased frequency of "red tokens." During text generation, green tokens are made to appear more often, while red tokens are suppressed, creating a detectable pattern

Distinguishing Watermarked Text: High z-score and a low p-value, as well as significantly more green tokens than red tokens, indicate the text is watermarked.

References

- Yuxian Gu, Li Dong, Furu Wei, Minlie Huang. 2023. MiniLLM: Knowledge Distillation of Large Language Models. *Proceedings of the IEEE Conference on Artificial Intelligence (AICoNf)*
- John Kirchenbauer, Jonas Geiping, Yuxin We. 2023. A Watermark for Large Language Models. *Proceedings of the ACM Conference on Advances in Neural Information Processing Systems (NeurIPS)*
- Byungjoo Kim, Suyoung Lee. 2023. Margin-based Neural Network Watermarking. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*
- Liang Chen, Yatao Bian, Yang Deng. 2024. WatME: Towards Lossless Watermarking Through Lexical Redundancy. *Proceedings of the Association for Computational Linguistics (ACL)*.

Method



- Training the Teacher Model:** We began by training the GPT-2 large model from OpenAI, which contains 774 million parameters, using a diverse set of input prompts.
- Generating Unwatermarked Logits (Dist. 1):** During this process, we produced the initial output, referred to as distribution 1, from the teacher model which consisted of a tensor of logits.
- Integrating Watermark:** We integrated a watermark into the model's fine-tuning process, enabling the generation of watermarked logits.
- Generating Watermarked Logits (Dist. 2):** Post watermark integration, the teacher model produced a new output distribution, known as distribution 2, distinguished by a different tensor of logits due to the watermark.
- Distilling the Teacher Model:** Using MiniLLM, we performed knowledge distillation from the teacher model (GPT-2 Large) to the student model (GPT-2 Large).
- Generating Student Model Distribution (Dist. 3):** After distillation, the watermarked distribution 2 served as the teacher model for further distillation, following the method from paper [1], to generate the final output distribution, termed distribution 3.
- Comparing Student Model Distribution (Dist. 3) to Teacher Model Distribution (Dist. 1):** To evaluate the extent to which the watermark was preserved throughout the distillation process, we conducted a detailed analysis using metrics such as the Fraction of Tokens Counted in the Greenlist, Z-score, and P-value. These indicators were employed to rigorously assess the efficacy of the watermark's preservation during distillation and to evaluate its reliability in the final model output. We will also conduct watermark detection, and further analysis using metrics such as the Rouge-L score and reverse Kullback-Leibler Divergence Score

Results

Distillation did NOT preserve the watermark

Result:

Tokens Counted (T)	# Tokens in Greenlist	Fraction of T in Greenlist	Z-score	P-value	Z-score Threshold	Prediction
75	19	25.3%	0.0667%	0.473	4.0	Human/Unwatermarked

Why isn't the watermark preserved?: Because the reverse KLD has p (teacher) following q (student), there are multiple possibilities for q to have a different variance. However, if KLD has q follow the distribution of p , there are fewer possibilities for different variances.

Performance Metrics of Teacher and Student Model

Model	#Params	Method	DollyEval (R-L)
GPT-2	774M	Teacher	11.9875
		KD	6.8652
		MiniLLM	10.485

New Algorithm

We propose a **joint optimization framework with regularization** to preserve the watermark through distillation

Proposed Formula to Preserve Watermark through Distillation

$$L_{total} = \alpha \cdot L_{task} + \beta \cdot L_{watermark} + \gamma \cdot L_{reg}$$

$$(1) \quad L_{task} = KL(p_{teacher}(y | x) || p_{student}(y | x))$$

$$(2) \quad L_{watermark} = Distance(W_{teacher}, W_{student})$$

$$(3) \quad L_{reg} = \lambda_1 \sum_j |w_j| + \lambda_2 \sum_j w_j^2$$

- (1) - Difference between watermarked outputs from the teacher and student models
- Measures how well the student model mimics the output distribution of the teacher model.

- (2) - KL divergence between the teacher and student model
- Measures how well the student model mimics the output distribution of the teacher model.

- (3) - Regularization term to prevent overfitting and to ensure the model remains generalizable.